Terna Engineering College
Computer Engineering Department

Class: TE **Sem.: VI**

Course: System Security Lab

PART A

(PART A : TO BE REFFERED BY STUDENTS)

Experiment No.05

A.1 Aim: Study the use of network reconnaissance tools like WHOIS, dig, traceroute, nslookup to gather information about networks and domain registrars.

A.2 Prerequisite:

1. Basic Knowledge of IP addresses, DNS.

A.3 Outcome:

After successful completion of this experiment students will be able to

Apply basic network command to gather basic network information.

A.4 Theory:

Network Reconnaissance:

- Act of reconnoitoring ---explore with the goal of finding something(especially to gain information about enemy)
- In the world of hacking, reconnaisance begins with "Footprinting"
- i.e accumulating data about target's environment, and finding vulnerabilities.
- Attacker gathers information in two phases viz: passive attacks and active attacks

Passive attacks

- Gathering information about a target without his/her knowledge….Eavesdropping
- Yahoo or google search
- Surfing online community groups
- Gathering information from websites of organisations. e.g. contact details, email address etc.
- Blogs, newsgroups, press releases etc.
- Going through job posting in particular job profiles

Reconnaissance Tools

- WHOIS, dig, traceroute, nslookup

1. **WHOIS:** WHOISistheLinuxutilityforsearchinganobjectinaWHOISdatabase.TheWHOIS database of a domain is the publicly displayed information about a domains ownership, billing, technical, administrative, and nameserver information. Running a WHOIS on your domain will look the domain up at the registrar for the domain information. All domains have WHOIS information. WHOIS database can be queried to obtain the following informationvia WHOIS:

   - Administrative contact details, including names, email addresses, and telephone numbers
   - Mailing addresses for office locations relating to the targetorganization
   - Details of authoritative name servers for each given domain

Example: Querying Facebook.com

ssc@ssc-OptiPlex-380:~$ whois facebook.com

For more information on Whois status codes, please visit https://www.icann.org/resources/pages/epp-status-codes-2014-06-16-en.

- Domain Name: facebook.com
- Registry Domain ID: 2320948_DOMAIN_COM-VRSN
- Registrar WHOIS Server: whois.markmonitor.com Registrar URL: http://www.markmonitor.com
- Updated Date: 2014-10-28T12:38:28-0700
- reation Date: 1997-03-28T21:00:00-0800
- Registrar Registration Expiration Date: 2020-03-29T21:00:00-0700 Registrar: MarkMonitor, Inc.

- Registrar IANA ID: 292
- Registrar Abuse Contact Email: abusecomplaints@markmonitor.com Registrar Abuse Contact Phone: +1.2083895740
- Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
- Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
- Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
- Registry Registrant ID:
- Registrant Name: Domain Administrator Registrant Organization: Facebook, Inc. Registrant Street: 1601 Willow Road, Registrant City: Menlo Park
- Registrant State/Province: CA Registrant Postal Code: 94025
- Registrant Country: US
- Registrant Phone: +1.6505434800
- Registrant PhoneExt:
- Registrant Fax: +1.6505434800
- Registrant Fax Ext:
- Registrant Email: domain@fb.com Registry Admin ID:
- Admin Name: Domain Administrator Admin Organization: Facebook, Inc. Admin Street: 1601 Willow Road, Admin City: Menlo Park
- Admin State/Province: CA Admin Postal Code: 94025 Admin Country: US
- Admin Phone: +1.6505434800 Admin Phone Ext:
- Admin Fax: +1.6505434800
- Admin Fax Ext:
- Admin Email: domain@fb.com Registry Tech ID:
- Tech Name: Domain Administrator

- Tech Organization: Facebook, Inc. Tech Street: 1601 Willow Road, Tech City: Menlo Park
- Tech State/Province: CA
- Tech Postal Code: 94025
- Tech Country: US
- Tech Phone: +1.6505434800
- Tech Phone Ext:
- Tech Fax: +1.6505434800
- Tech Fax Ext:
- Tech Email: domain@fb.com
- Name Server: b.ns.facebook.com
- Name Server: a.ns.facebook.com
- DNSSEC: unsigned
  URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/

  >>> Last update of WHOIS database: 2015-07-16T21:08:30-0700 <<<

The Data in MarkMonitor.com's WHOIS database is provided by MarkMonitor.com for information purposes, and to assist persons in obtaining information about orrelated to a domain name registration record. MarkMonitor.com does not guarantee its accuracy. By submitting a WHOIS query, you agree that you will use this Data only for lawful purposes and that, under no circumstances will you use this Data to:

(1) allow, enable, or otherwise support the transmission of mass unsolicited, commercial advertising or solicitations via e-mail (spam);or
(2) Enable high volume, automated, electronic processes that apply to MarkMonitor.com (or its systems).

MarkMonitor.com reserves the right to modify these terms at any time. By submitting this query, you agree to abide by this policy.

MarkMonitor is the Global Leader in Online Brand Protection. MarkMonitor Domain Management(TM)

MarkMonitor Brand Protection(TM) MarkMonitorAntiPiracy(TM) MarkMonitorAntiFraud(TM) Professional and Managed Services

Visit MarkMonitor at http://www.markmonitor.com Contact us at +1.8007459229

In Europe, at +44.02032062220 ssc@ssc-OptiPlex-380:~$

2. **Dig** - Dig is a networking tool that can query DNS servers for information. It can be veryhelpful fordiagnosingproblemswithdomainpointingandisagoodwaytoverifythatyourconfiguration is working. The most basic way to use dig is to specify the domain we wish toquery:

Example:

$ dig duckduckgo.com

3. **Traceroute** - traceroute prints the route that packets take to a network host. Traceroute utility uses the TTL field in the IP header to achieve its operation. For users who are new to TTL field, this field describes how much hops a particular packet will take while traveling on network. So, thiseffectivelyoutlinesthelifetimeofthepacketonnetwork.Thisfieldisusuallysetto 32or64. Each time the packet is held on an intermediate router, it decreases the TTL value by 1. When a router finds the TTL value of 1 in a received packet then that packet is not forwarded but instead discarded.Afterdiscardingthepacket,routersendsanICMPerrormessageof―Timeexceeded‖ back to the source from where packet generated. The ICMP packet that is sent back contains the IP address of the router. So now it can be easily understood that traceroute operates by sending packets with TTL value starting from 1 and then incrementing by one each time. Each time a routerreceivesthepacket,itcheckstheTTLfield,ifTTLfieldis1thenitdiscardsthepacketand sends the ICMP error packet containing its IP address and this is what traceroute requires. So traceroute incrementally fetches the IP of all the routers between the source and thedestination.

Example:

$traceroute example.com

PART B

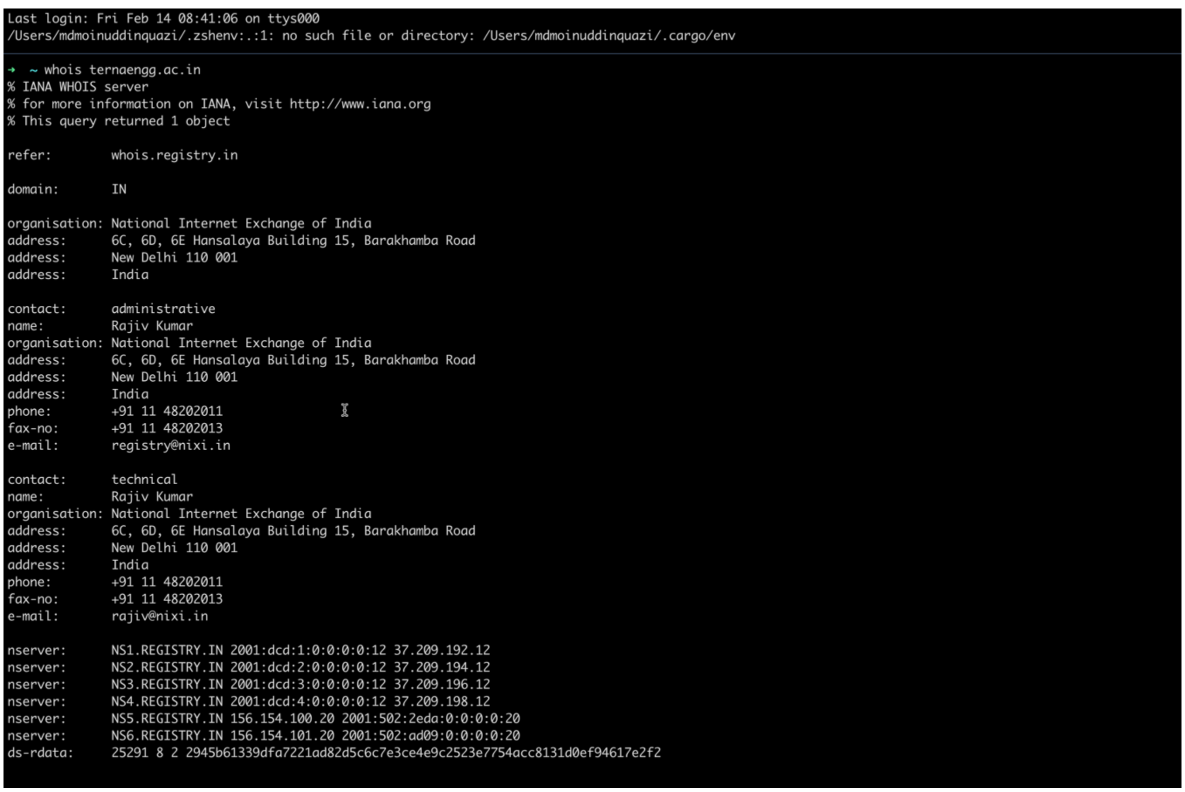
(PART B : TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)

Roll No.C54	**Name:Quazi Md Moinuddin**
Class :TE-C	**Batch :C3**
Date of Experiment: 6-2-25	**Date of Submission:15-2-25**
Grade :	

B.1 Output of Reconnaissance Tools

(add snapshot of output of Reconnaissance Tools)



```
Last login: Fri Feb 14 08:41:06 on ttys000
/Users/mdmoinuddinquazi/.zshenv:.:1: no such file or directory: /Users/mdmoinuddinquazi/.cargo/env

➜  ~ whois ternaengg.ac.in
% IANA WHOIS server
% for more information on IANA, visit http://www.iana.org
% This query returned 1 object

refer:        whois.registry.in

domain:       IN

organisation: National Internet Exchange of India
address:      6C, 6D, 6E Hansalaya Building 15, Barakhamba Road
address:      New Delhi 110 001
address:      India

contact:      administrative
name:         Rajiv Kumar
organisation: National Internet Exchange of India
address:      6C, 6D, 6E Hansalaya Building 15, Barakhamba Road
address:      New Delhi 110 001
address:      India
phone:        +91 11 48202011
fax-no:       +91 11 48202013
e-mail:       registry@nixi.in

contact:      technical
name:         Rajiv Kumar
organisation: National Internet Exchange of India
address:      6C, 6D, 6E Hansalaya Building 15, Barakhamba Road
address:      New Delhi 110 001
address:      India
phone:        +91 11 48202011
fax-no:       +91 11 48202013
e-mail:       rajiv@nixi.in

nserver:      NS1.REGISTRY.IN 2001:dcd:1:0:0:0:0:12 37.209.192.12
nserver:      NS2.REGISTRY.IN 2001:dcd:2:0:0:0:0:12 37.209.194.12
nserver:      NS3.REGISTRY.IN 2001:dcd:3:0:0:0:0:12 37.209.196.12
nserver:      NS4.REGISTRY.IN 2001:dcd:4:0:0:0:0:12 37.209.198.12
nserver:      NS5.REGISTRY.IN 156.154.100.20 2001:502:2eda:0:0:0:0:20
nserver:      NS6.REGISTRY.IN 156.154.101.20 2001:502:ad09:0:0:0:0:20
ds-rdata:     25291 8 2 2945b61339dfa7221ad82d5c6c7e3ce4e9c2523e7754acc8131d0ef94617e2f2
```

```
33  * * *
34  * * *
35  * * *
36  * * *
37  * * *
38  * * *
39  * * *
40  * *^C

→  ~ traceroute ternaengg.ac.in
traceroute to ternaengg.ac.in (195.200.9.131), 64 hops max, 40 byte packets
 1  192.168.230.107 (192.168.230.107)  6.705 ms  6.107 ms  5.105 ms
 2  255.0.0.0 (255.0.0.0)  34.619 ms  48.432 ms  33.655 ms
 3  255.0.0.2 (255.0.0.2)  33.896 ms  44.261 ms  41.035 ms
 4  255.0.0.3 (255.0.0.3)  40.873 ms  30.813 ms  46.535 ms
 5  172.17.178.2 (172.17.178.2)  34.061 ms
    172.17.178.3 (172.17.178.3)  38.977 ms
    172.17.178.2 (172.17.178.2)  38.376 ms
 6  192.168.217.82 (192.168.217.82)  46.411 ms  43.969 ms
    192.168.217.88 (192.168.217.88)  34.532 ms
 7  * * *
 8  * * *
 9  * 103.198.140.174 (103.198.140.174)  42.783 ms *
10  49.44.220.243 (49.44.220.243)  110.267 ms  32.324 ms
    103.198.140.174 (103.198.140.174)  47.665 ms
11  195.66.227.67 (195.66.227.67)  309.057 ms
    103.198.140.43 (103.198.140.43)  185.829 ms
    103.198.140.105 (103.198.140.105)  167.413 ms
12  po50.er1.thn1.net.ans.uk (78.24.91.30)  176.365 ms  172.920 ms  191.624 ms
13  195.66.227.67 (195.66.227.67)  157.240 ms
    po50.er1.thn1.net.ans.uk (78.24.91.30)  187.712 ms  170.993 ms
14  po20.er1.man8.net.ans.uk (78.24.91.32)  160.293 ms  220.389 ms  392.259 ms
15  po10.er2.man8.net.ans.uk (78.24.91.29)  307.125 ms  526.709 ms
    po30.er2.man6.net.ans.uk (78.24.91.1)  314.450 ms
16  po30.er2.man6.net.ans.uk (78.24.91.1)  443.317 ms  349.764 ms  412.221 ms
17  * 153.92.2.242 (153.92.2.242)  169.366 ms
    po30.er2.man6.net.ans.uk (78.24.91.1)  175.021 ms
18  153.92.2.242 (153.92.2.242)  334.141 ms  412.781 ms  198.557 ms
19  * 153.92.2.242 (153.92.2.242)  321.695 ms *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  *
```

```
11  195.66.227.67 (195.66.227.67)  309.057 ms
    103.198.140.43 (103.198.140.43)  185.829 ms
    103.198.140.105 (103.198.140.105)  167.413 ms
12  po50.er1.thn1.net.ans.uk (78.24.91.30)  176.365 ms  172.920 ms  191.624 ms
13  195.66.227.67 (195.66.227.67)  157.240 ms
    po50.er1.thn1.net.ans.uk (78.24.91.30)  187.712 ms  170.993 ms
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    po30.er2.man6.net.ans.uk (78.24.91.1)  314.450 ms
16  po30.er2.man6.net.ans.uk (78.24.91.1)  443.317 ms  349.764 ms  412.221 ms
17  * 153.92.2.242 (153.92.2.242)  169.366 ms
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19  * 153.92.2.242 (153.92.2.242)  321.695 ms *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
^C

→  ~ dig ternaengg.ac.in

; <<>> DiG 9.10.6 <<>> ternaengg.ac.in
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 54981
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;ternaengg.ac.in.               IN      A

;; ANSWER SECTION:
ternaengg.ac.in.        60      IN      A       195.200.9.96

;; Query time: 186 msec
;; SERVER: 192.168.230.107#53(192.168.230.107)
;; WHEN: Sat Feb 15 21:30:25 IST 2025
;; MSG SIZE  rcvd: 60

→  ~
```

```
16  po30.er2.man6.net.ans.uk (78.24.91.1)  443.317 ms  349.764 ms  412.221 ms
17  * 153.92.2.242 (153.92.2.242)  169.366 ms
    po30.er2.man6.net.ans.uk (78.24.91.1)  175.021 ms
18  153.92.2.242 (153.92.2.242)  334.141 ms  412.781 ms  198.557 ms
19  * 153.92.2.242 (153.92.2.242)  321.695 ms *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
^C

→  ~ dig ternaengg.ac.in

; <<>> DiG 9.10.6 <<>> ternaengg.ac.in
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 54981
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;ternaengg.ac.in.               IN      A

;; ANSWER SECTION:
ternaengg.ac.in.        60      IN      A       195.200.9.96

;; Query time: 186 msec
;; SERVER: 192.168.230.107#53(192.168.230.107)
;; WHEN: Sat Feb 15 21:30:25 IST 2025
;; MSG SIZE  rcvd: 60

→  ~ nslookup ternaengg.ac.in
Server:         2409:40c0:1005:babe::c2
Address:        2409:40c0:1005:babe::c2#53

Non-authoritative answer:
Name:   ternaengg.ac.in
Address: 195.200.9.96

→  ~
```

B.2 Commands / tools used with syntax:

•

- ➜ **~ whois ternaengg.ac.in**
- ➜ **~ traceroute ternaengg.ac.in**
- ➜ **~dig ternaengg.ac.in**
- ➜ **~nslookup ternaengg.ac.in**

B.3 Question of Curiosity: *(At least 3 questions should be handwritten)*

1. What information is grabbed from Whois?
2. What information is grabbed from traceroute?
3. What information is grabbed from dig?
4. After using traceroute how attacker can use the information, based on the same what kind of attacks can be applied?
5. How does nslookup help in diagnosing DNS-related issues?
6. What are the ethical considerations when using network reconnaissance tools like WHOIS and traceroute?

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CSS Expt 5

1] What information is grabbed from WHOIS?
→ WhoIS is protocol used to retrieve info about domain names, IP addresses, & related entities, The info" that can be retrieved from a WHO is query includes:
- Domain name
- Registrant information
- Domain ~~Cro~~ Creation & Expiration Dates
- Name servers
- Administrative & Technical Contact

2] What information is grabbed from traceroute?
→ Traceroute is a tool used to trace the path the packets take from a source machine to a destinat It provides the following info".
- Hop Count
- Router IP Addresses
- Round-Trip Time (RTT)
- Route Path

3] What information is grabbed from dig?
→ dig (Domain Information Groper) is a command-tool used for querying DNS records. It retrieves various DNS-related information, including:
- A Records: IP address with domain name.
- MX Records: Mail exchange servers responsible for recieving emails for the domain
- NS Records: The name servers for the domain
- CNAME Records, TXT Records, SOA Records.

B.4 Conclusion:

(Write appropriate conclusion.)

Network reconnaissance tools like WHOIS, traceroute, dig, and nslookup are valuable for diagnosing network issues, troubleshooting DNS problems, and gathering information about domain configurations.